# SENG202 – Software Engineering Project Workshop

## Lecture 5 – Expectations for deliverable 1

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# Announcements

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- Halfway through iteration 1 already … 2 weeks left!!!
- Tidy up your activity logs
  - Volume is increasing
  - Logs seem more consistent with expectations
  - Better! … But very imbalanced.
- First weekly reflection due today at midday
  - You have, now, enough information to start reflecting about your achievements, your team's achievements, etc.



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# Overview



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# Deliverables - assessed - 15%

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- Project setup checklist
- Reflections and activity logs
- Project documentation – 80%
- Initial increment (source code and JUnit tests) – 10%
- Presentation – 10%






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# Objectives of this lecture

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- Provide general feedback on presentations so far (short)
- Give indications on how to approach, focused on the most important aspects (weights and based on FAQ):
  - Stakeholders
  - System context
  - Use case diagram
  - Acceptance tests
  - Initial class diagram
  - Deployment diagram
  - Planning
  - Presentation
- Answer questions you may have (short)



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# General feedback



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# Progress after week 2 was, mostly, as expected

- Some very good ideas! … Not bad I should say…
- Heterogeneous progress
  - Some of you have focused on the scope/market but not much on the requirements/design/diagrams
    - need to translate the ideas into proper descriptions
  - Some of you have focused on the requirements/design/diagrams but not much on the scope/market
    - need to be more specific and prioritize
  - Some of you seem to have explored lots of options
    - need to catch up on the other aspects

- Some of the projects seem to be too ambitious
  - Prioritize, focus on your unique selling point first, even if you are targeting a niche

- Something we observed during the craft activity
  - Teams spending a lot of time on initial discussions
  - Teams not communicating enough or frequently enough
  - Great dedication: everyone supported the team effort and wanted to finish the craf even if it required to stay a bit longer…



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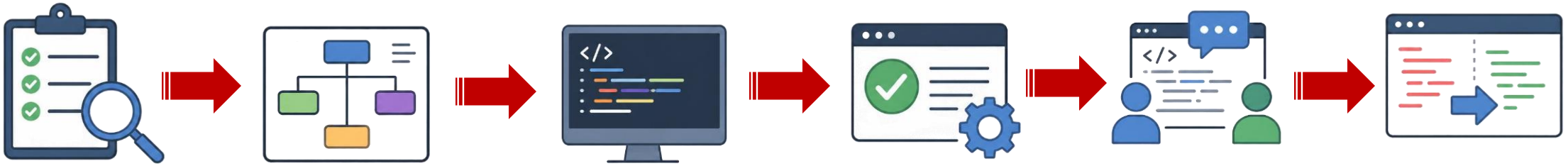
# Development process definition



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# Software Engineering activities and sequential processes

- The waterfall model as it has **never existed, but everyone refers to**

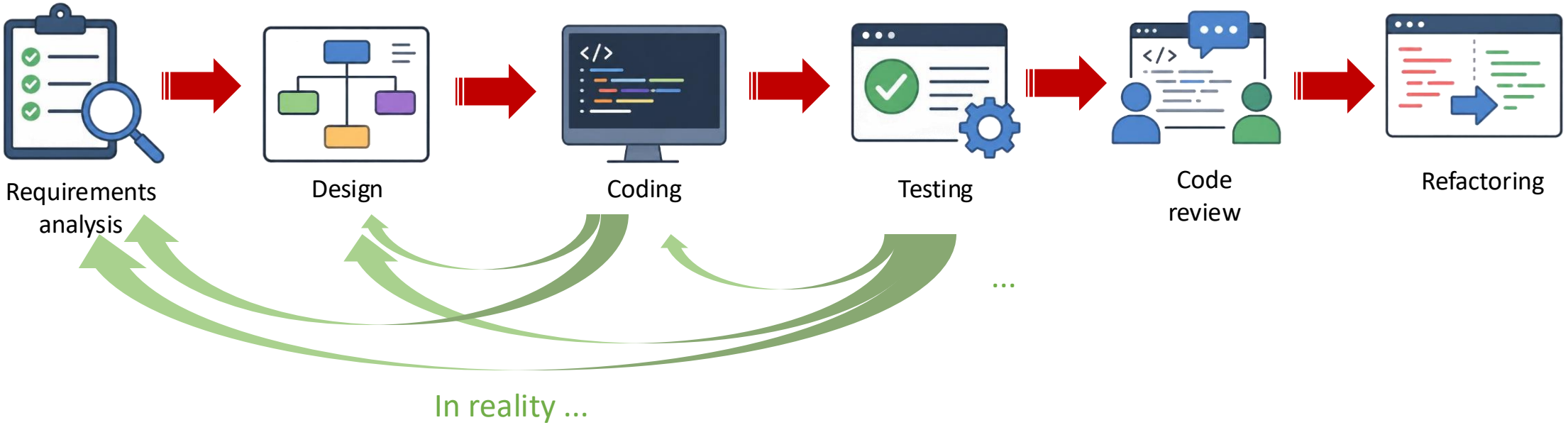





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# Software Engineering activities and sequential processes

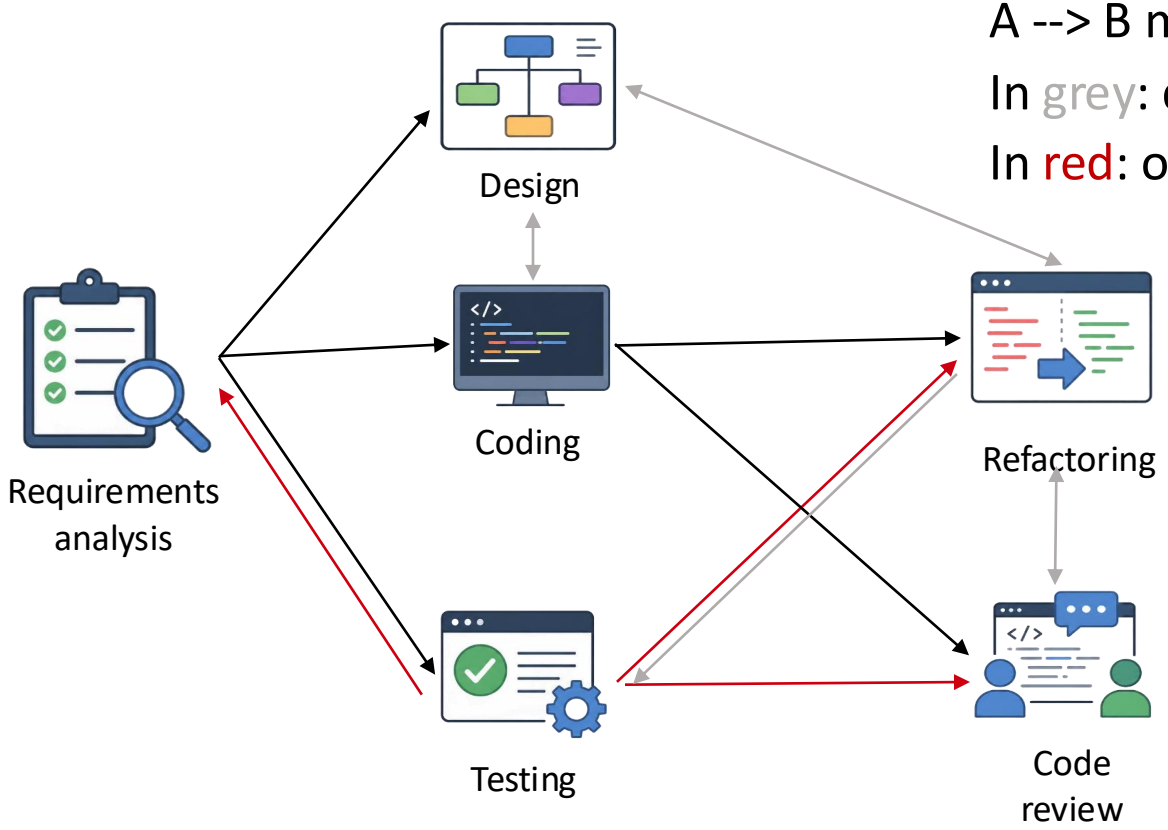
- The waterfall model as it has **existed in practice, but no one refers to**






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# A high-level view on Software Engineering activities (dependencies)




A --> B means B depends on A (A preceeds B, to some extent)

In grey: depend on the context (e.g. iteration)

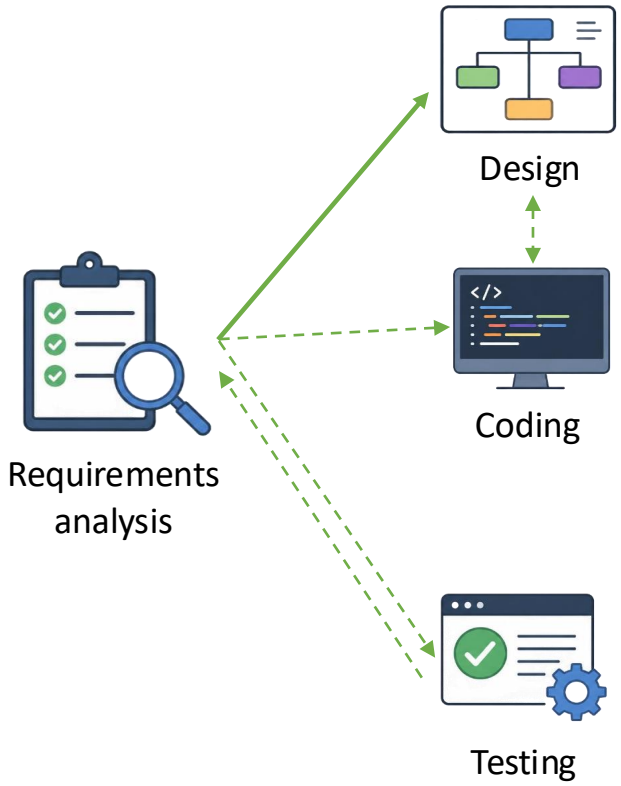
In red: often neglected by juniors but very important

- Quite complicated
- Topological sort impossible (cycles)
- Hence the importance of iterative and incremental approaches. SENG202 focuses on one approach (spiral) while SENG302 focuses on another one (Agile).
- Practice in industry is very heterogeneous/hybrid.



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# A high-level view on Software Engineering activities (iteration 1)






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# Project documentation

Stakeholders



Requirements analysis



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# Stakeholders

- Who will be using your product?
- Who will be affected or who will impact
  - Product
  - Development process
- Who is commissioning the project?
- People, roles, organizations, etc.
- Need to be prioritized
  - Use basic scale (low, medium, high)



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# Be specific - Why does it matter?

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- Helps refining the requirements …
  - It is difficult to make everyone happy
  - It is difficult to identify needs when a stakeholder is 'government'

- Will have impact on the design and quality of the system
  - Graphical interface
  - Use cases and Features definition and priority
  - Example
    - Responsibility: are your stakeholders major (i.e. above 20 years old)?
    - Response time: are your stakeholders using the app under time pressure?



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# Stakeholders - example

- Professional athletes, casual exercisers (including their relatives)
- Runners, cyclists, swimmers
- Personal trainers, sport teams
- Medical specialists
- Data analysts
- Retailers, sporting goods/fitness equipment manufacturers
- Social media
- UC



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# Stakeholders concerns – examples

- Professional athletes, casual exercisers (including their relatives)
  - Ability to track fitness progress
- Personal trainers , sport teams
  - Responsive and reliable application, intuitive interface
- Medical specialists
  - Collect and analyze data to provide better health treatment
- Data analysts
  - Relevant analyses, informative data visualization
  - Consumer research, better marketing
- Retailers, Sporting goods/fitness equipment manufacturers
  - Increased sales of complementary fitness goods
- Social media
  - Increased user retention
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  - Reputation
  - Education and research quality



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# Let's take an example

- SENG202-25 – application for trampers in NZ (trip planning, huts and campsites, ratings, souvenirs, gear list)

- Name of the proposed app: TrampSafe

- Stakeholders (sorted by importance) - and concerns
  - LandSAR NZ – more specifically the search management team, who need precise information when starting a SAR operation
  - Intermediate to advanced trampers – who understand the importance of being located rapidly in case of incident
  - PLB providers – trampers who want to facilitate access to these devices
  - DOC – provide data and alert trampers if conditions are dangerous
  - Tramping gear retailers – who could advertise for their gears and stores location via the app
  - Convenience stores selling tramping specific food – advertisement, increase traffic, inform trampers of shortages



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# Project documentation

System context



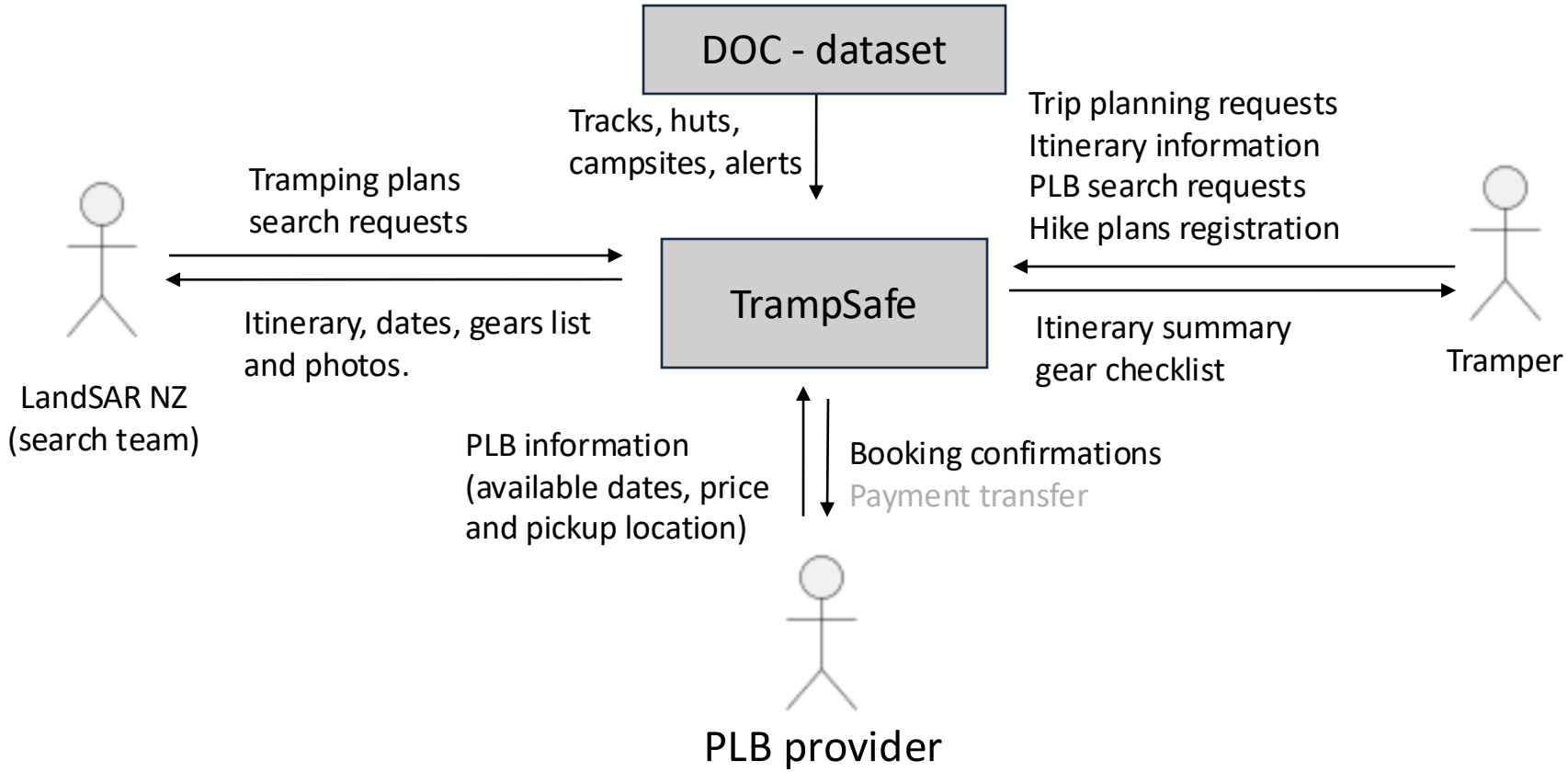
Requirements analysis



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# What is a system context diagram?

A high-level (black-box) representation of the system, its boundaries, and everything that interacts with it. Focus on main stakeholders and data exchange.






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# Project documentation

## Use cases



Requirements analysis



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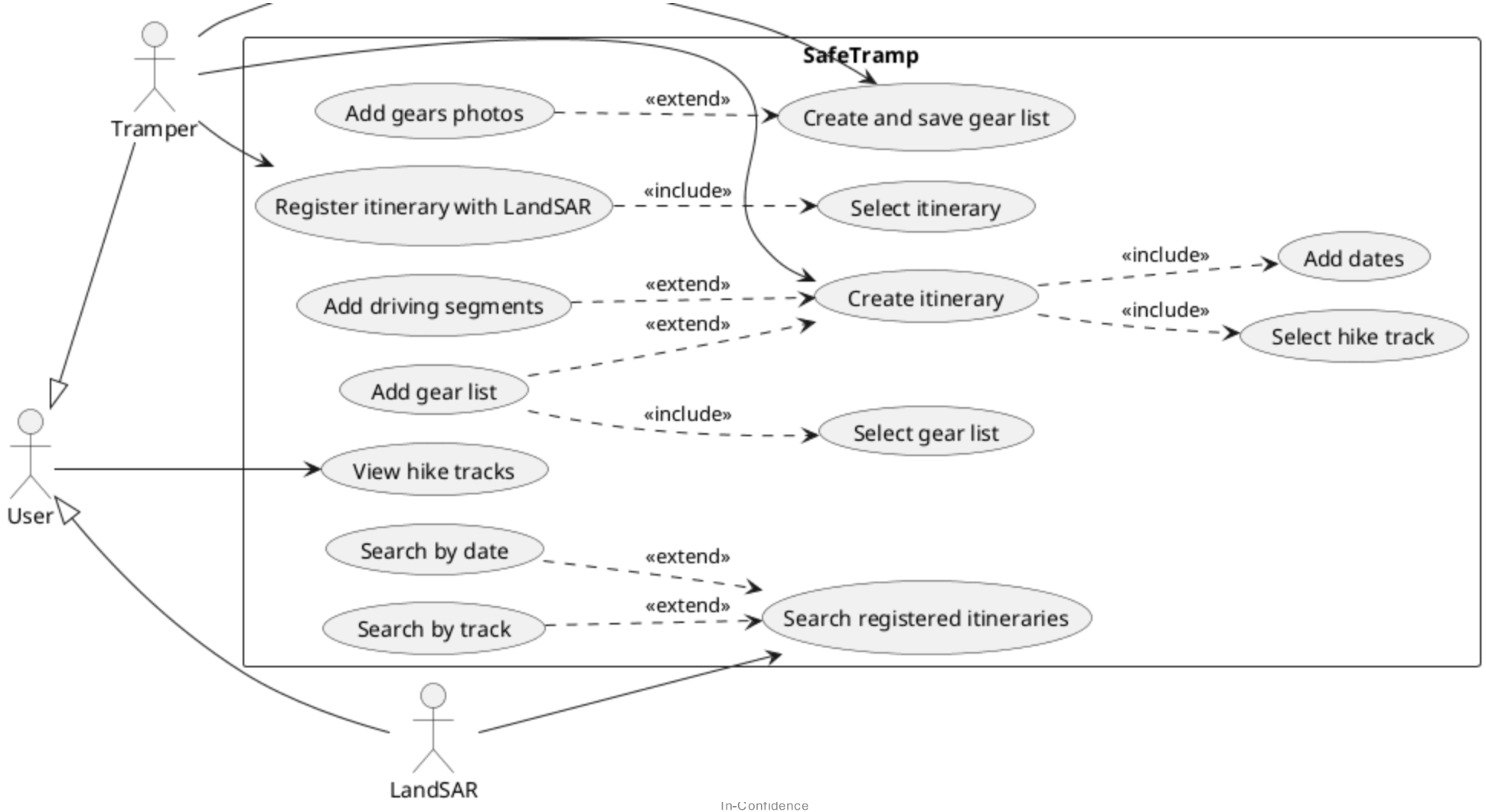
# A software engineer with a customer...






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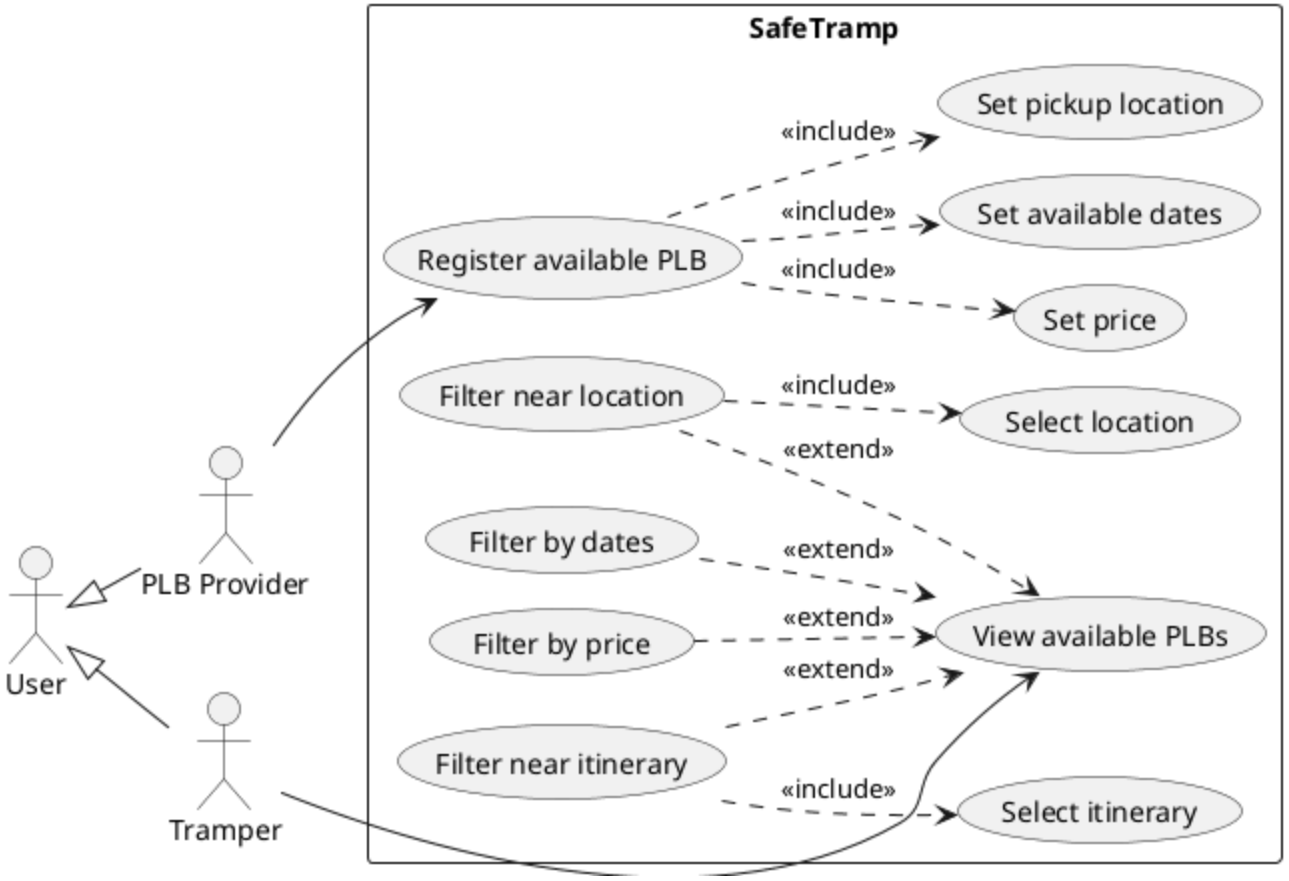
# Use case diagram (MVP)






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# Use case diagram (extension for the final product)






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# Priorities

- Use cases must be prioritized
  - Consider dependencies (includes/extends)

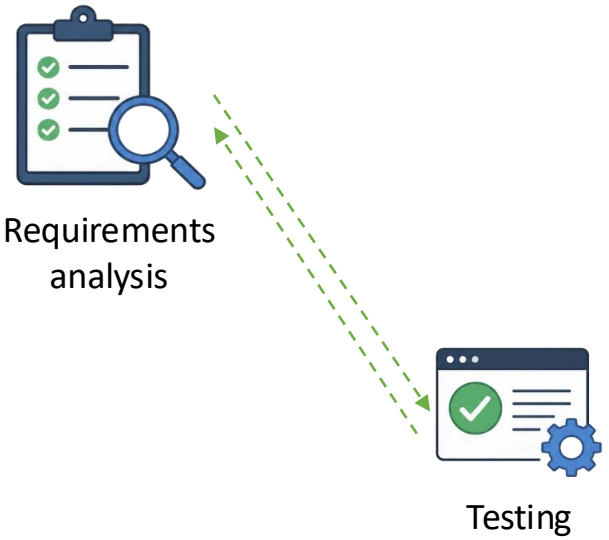
- High priority use cases must be described with pre-condition / scenarios / post-conditions



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# Project documentation

Acceptance tests






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# Given [precondition(s)] when [action] then [postcondition]

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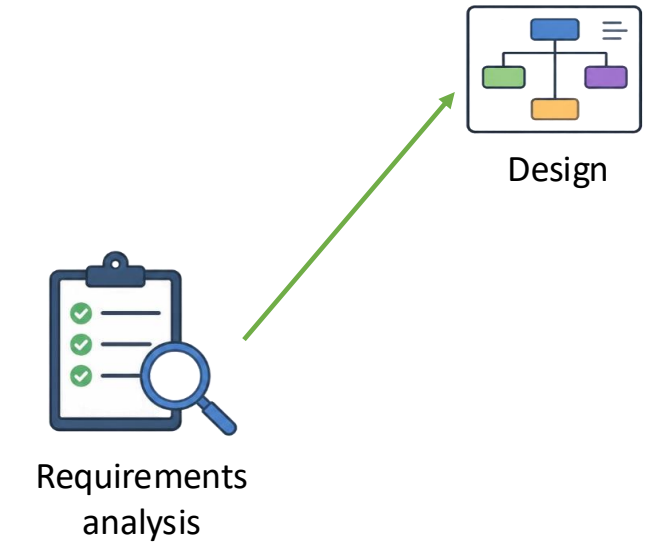
- Should cover high priority use case scenarios
- Should be atomic (one business rule or one observable behaviour)
- Example: Register Itinerary With LandSAR
  - **Given** a tramper has a saved an itinerary **When** the tramper registers the itinerary with LandSAR **Then** the itinerary becomes visible to LandSAR
  - **Given** a tramper has not saved any itinerary **When** the tramper attempts to register an itinerary with LandSAR **Then** the system prevents the registration And the system informs the tramper that a saved itinerary is required



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# Project documentation

Initial class diagram + deployment diagram?






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# Boundary/Control/Entity

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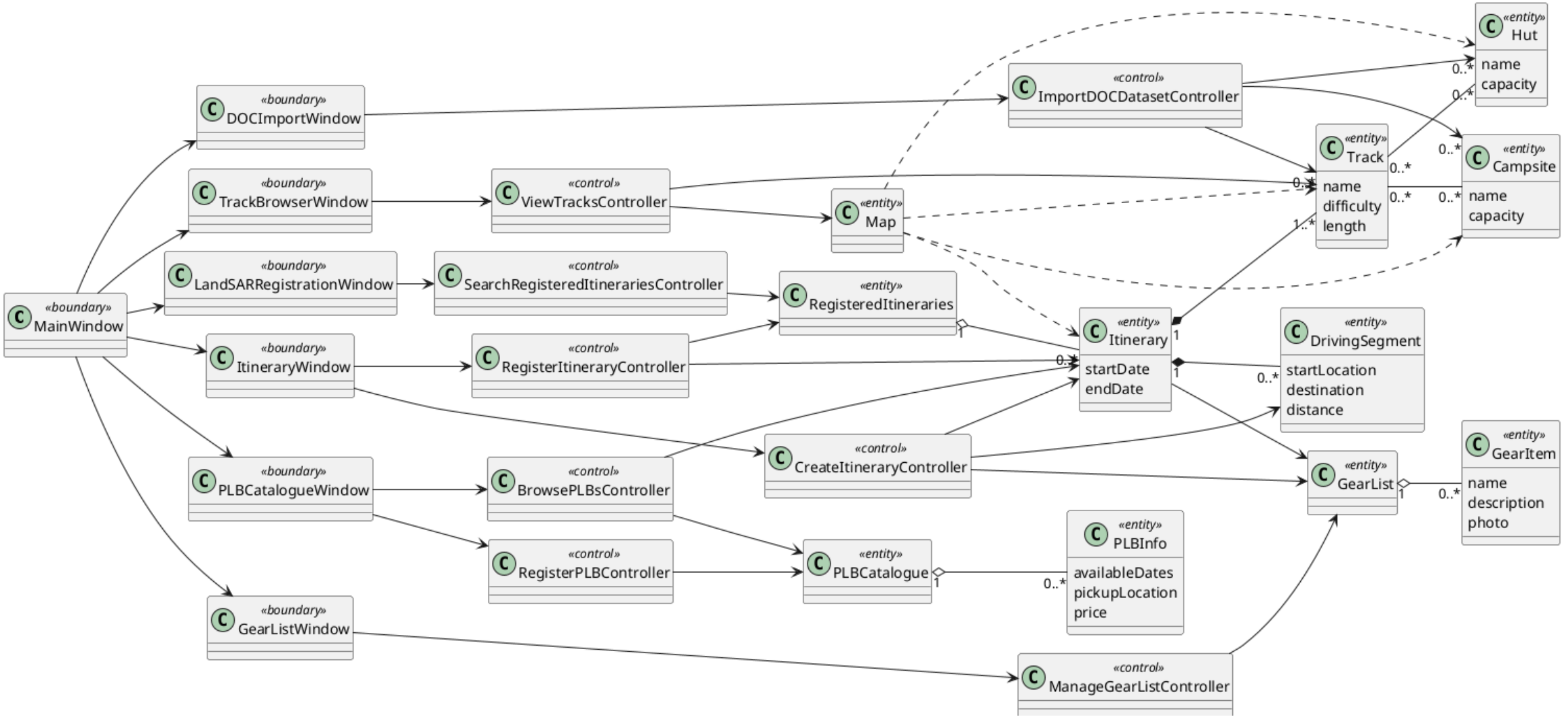
- For your initial class diagram, consider using UML Boundary/Control/Entity stereotypes
  - Boundary: interfaces of the system with users/external systems
  - Control: processing classes, where methods are expected to do some computation
  - Entity: data structure

- Boundary classes easily map to the actors/APIs...
- Control classes easily map to use cases
- Entity classes easily map to concepts in the domain knowledge

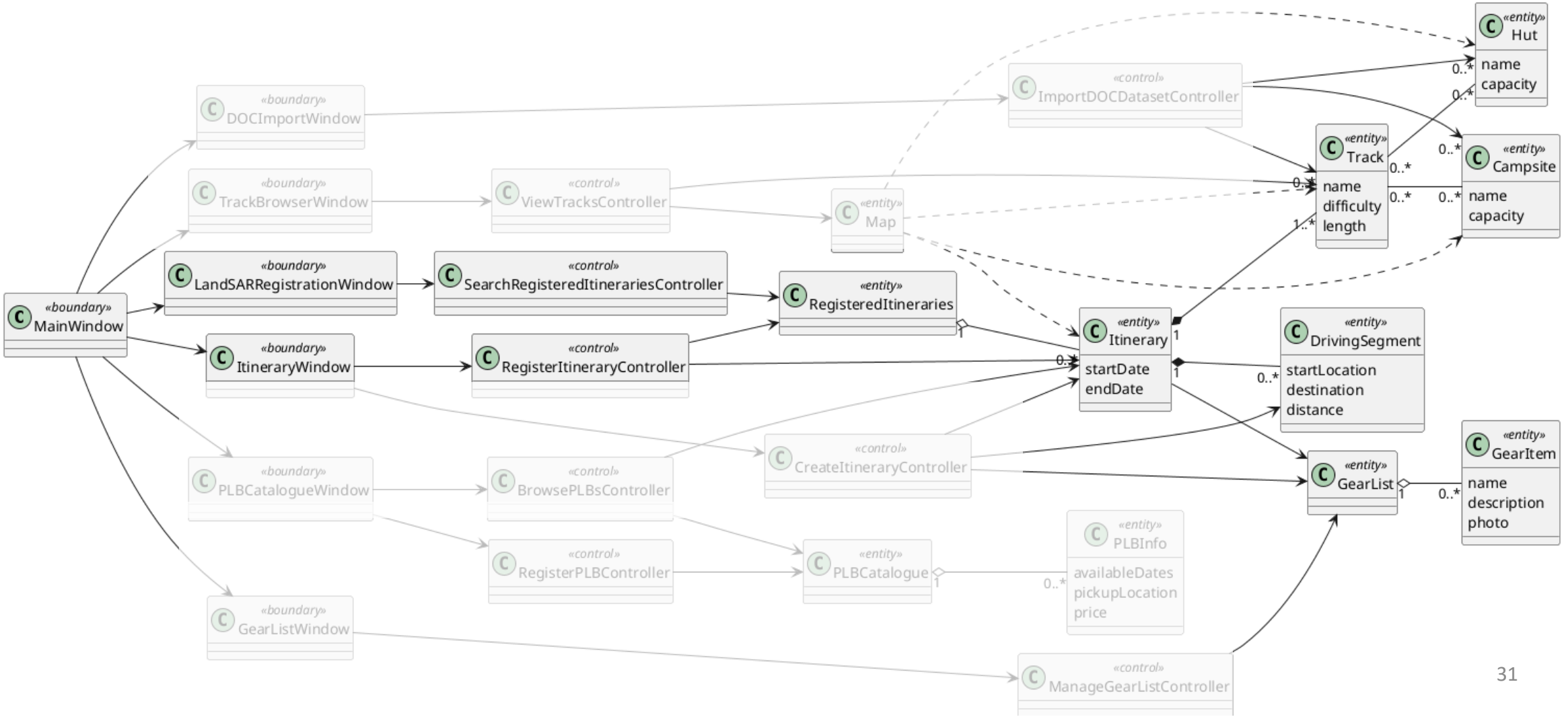


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# Initial Class Diagram



# Initial Class Diagram



# Deployment diagram

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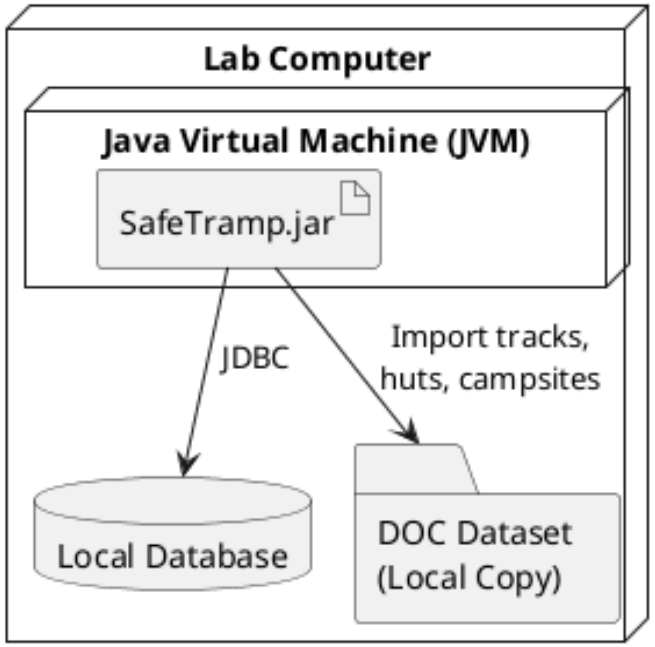
- High-level UML deployment diagram
- Physical deployment of software artifacts on nodes
  - Nodes (physical hardware to deploy system)
    - Monitor
    - Modem
    - Servers (e.g., caching, application, DB)
    - Clients
    - Etc.
  - Relationships
    - Important physical link between nodes (e.g., TCP/IP, Ethernet, JDBC, REST, RMI)
  - Software artifacts
    - Execs, binary, archive, DB, source, libs, packages, files, documents, scripts, etc.



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# Deployment Diagram - example

- Yours is likely going to be more complete and slightly more complicated … But aim at Keeping It Simple






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# Project documentation

Planning



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# Planning - Hofstadter's Law

"It always takes longer than you expect, even when you take into account Hofstadter's Law."



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# Presentation



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# Expected content

- What is the system context and business vision/rationale
- Who are your stakeholders and concerns
- What are main use cases (no detailed requirements), with GUI low fidelity prototypes
- What are the key drivers
- Example (major) acceptance tests  (optional)
- Overview of UML class diagram (patterns, layers, packages, etc.)
- What are main risks (what could go wrong; how is it mitigated)
- Overview of project plan, how will goal be achieved (optional)
- …



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# Wrapping up … Questions?



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